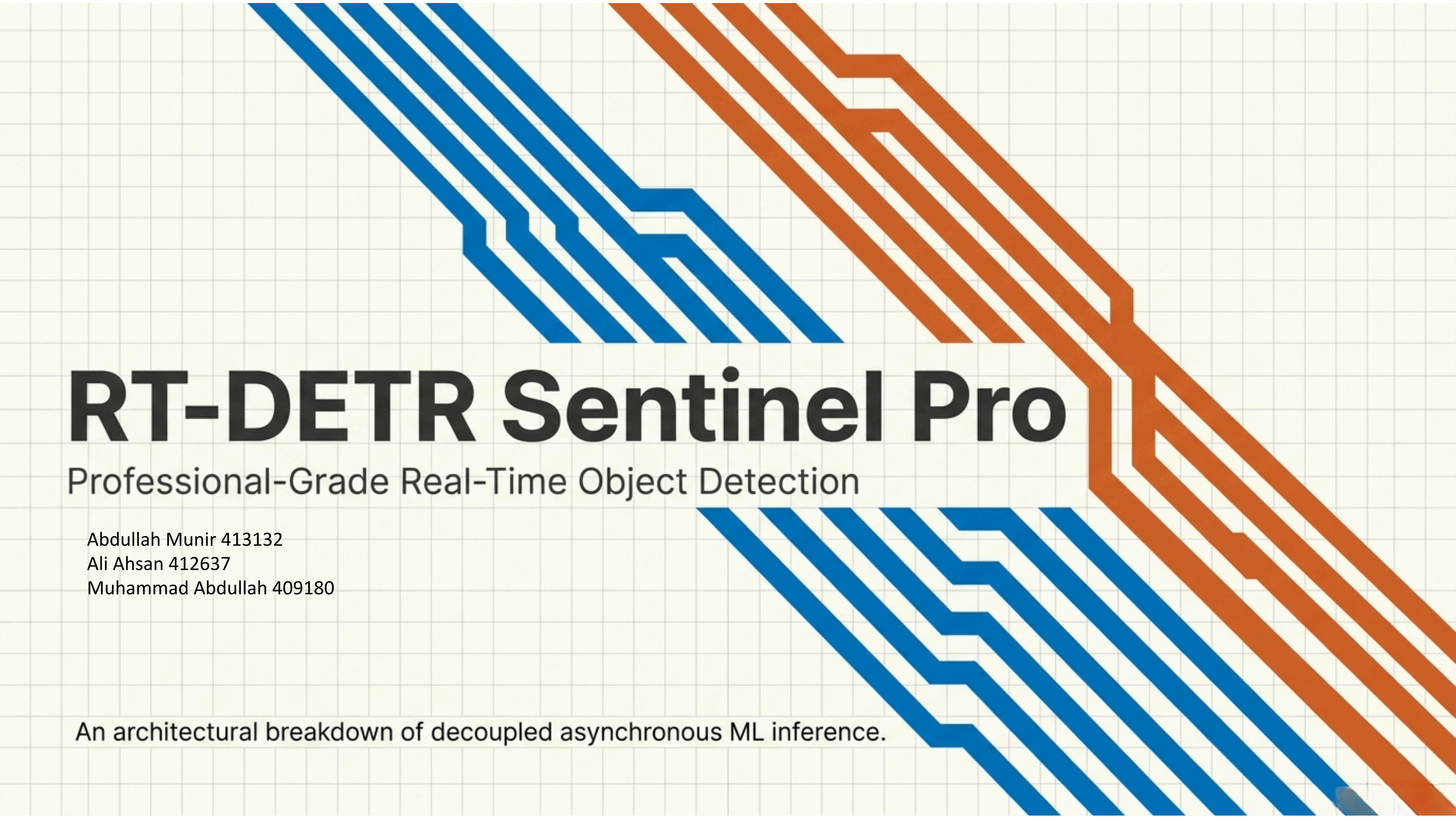




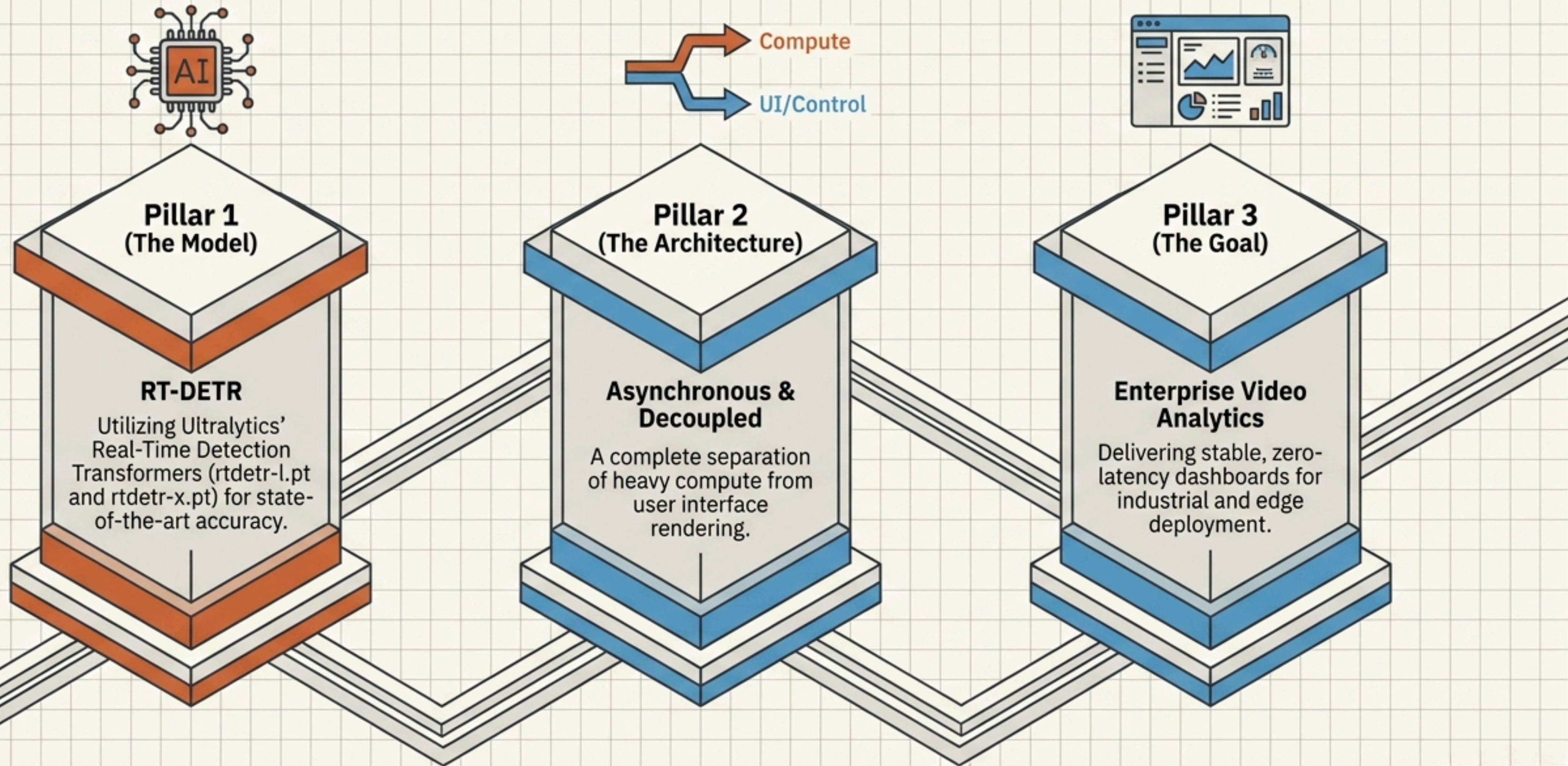
RT-DETR Sentinel Pro

Professional-Grade Real-Time Object Detection

Abdullah Munir 413132
Ali Ahsan 412637
Muhammad Abdullah 409180

An architectural breakdown of decoupled asynchronous ML inference.

Bridging SOTA Transformers with real-time enterprise analytics

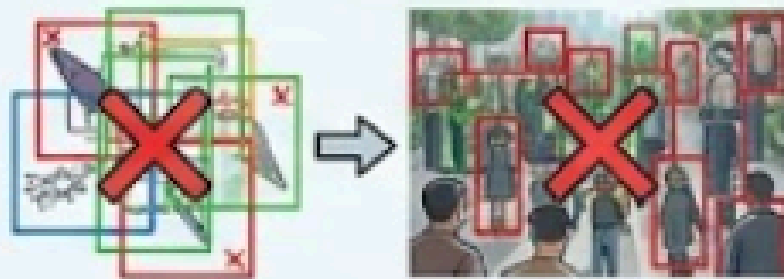


Why Not Other Models?

Limitations of Alternatives

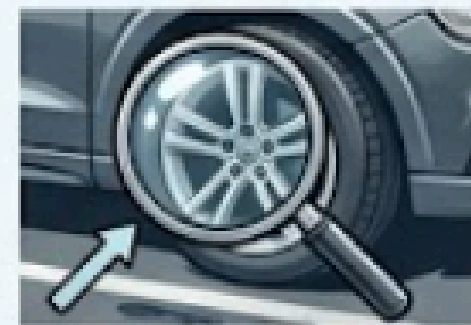
YOLO (You Only Look Once)

Requires NMS
(Non-Maximum Suppression)



Cluttered object detection

Local Reasoning



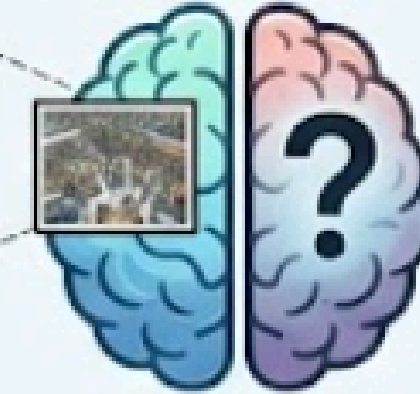
Local Reasoning

Other CNNs (Convolutional Neural Networks)

Lack Global Context



Local Features

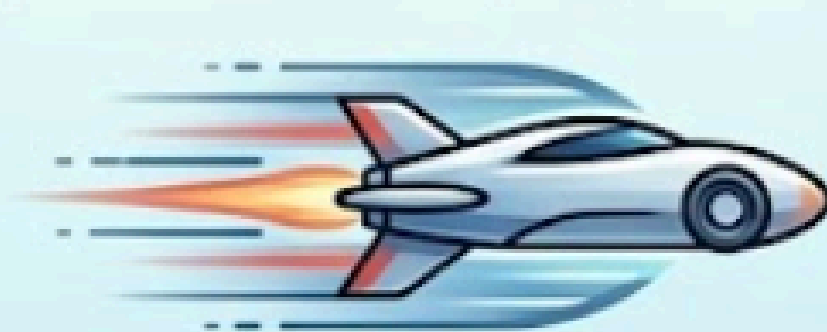


disconnected whole

The Solution:

RT-DETR (Real-Time Detection Transformer)

Speed & Accuracy: High Real-Time Performance & Top Precision



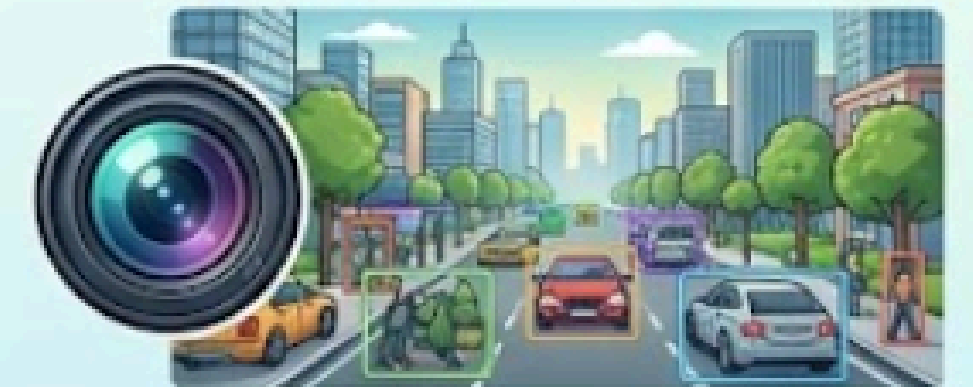
✓ Fast & Accurate
Detection



✓ Fast & Accurate
Detection



Global Reasoning
(Self-Attention)



Global Context for a
Complete Understanding

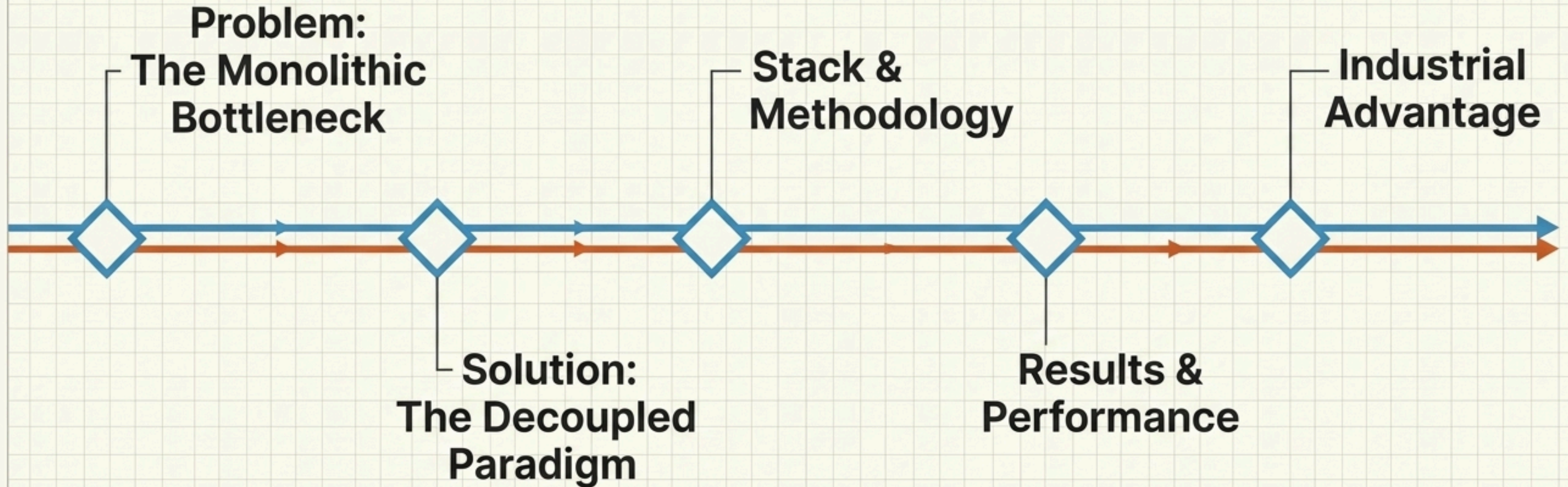


SPEED

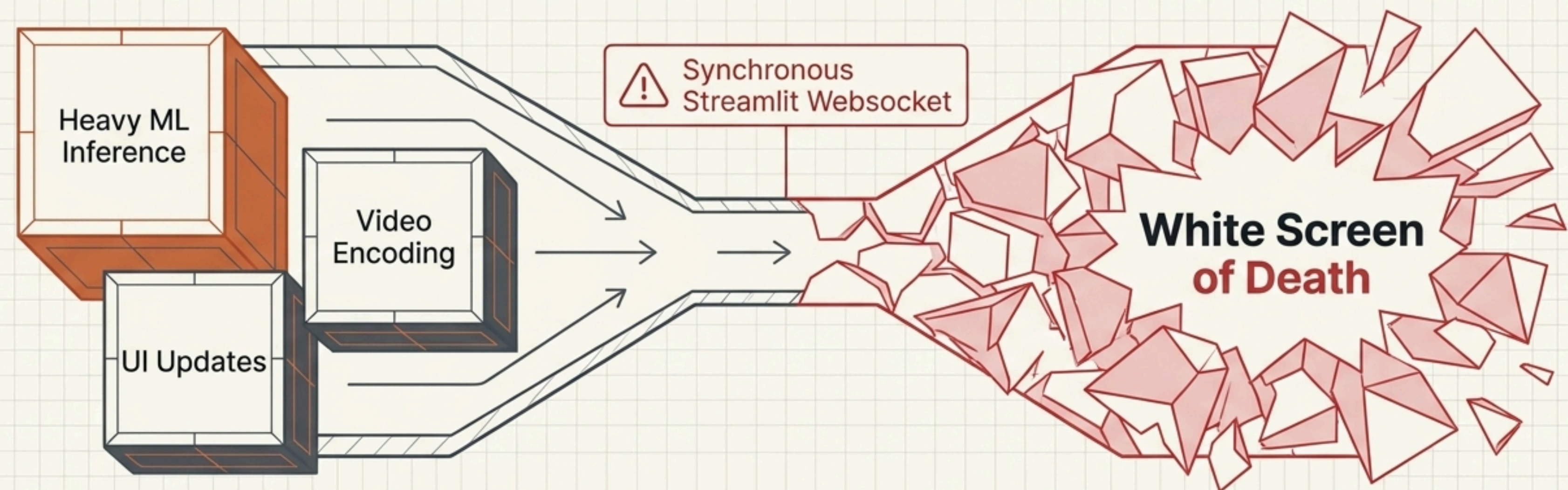


ACCURACY

The Engineering Pipeline



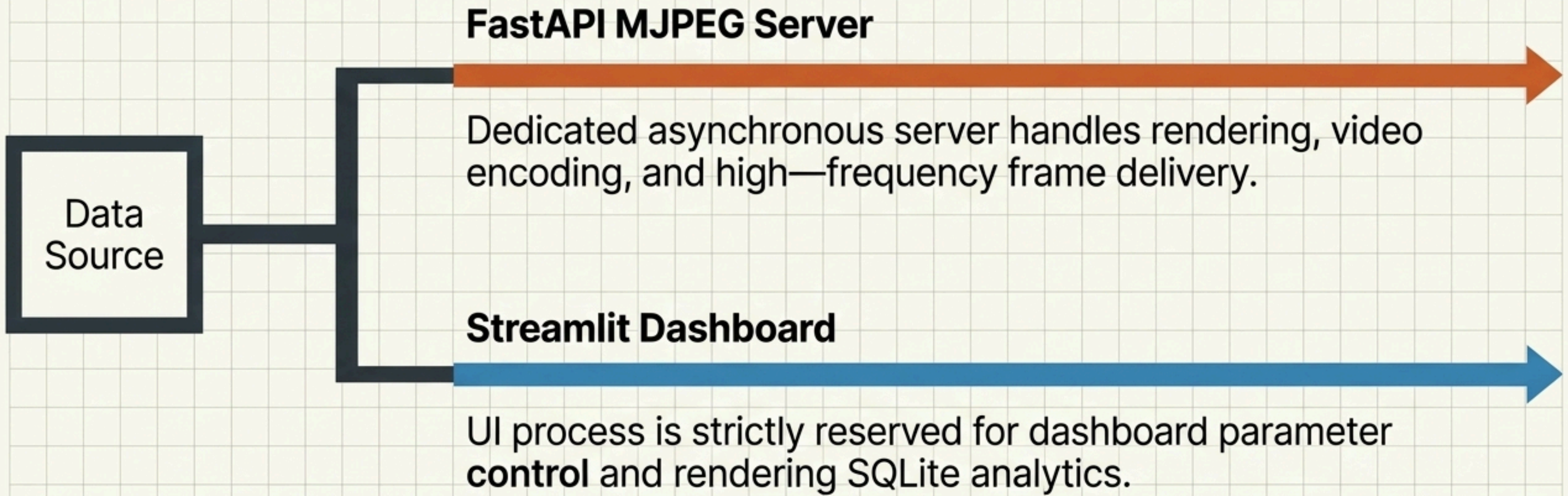
Synchronous architectures collapse under heavy inference loads



Core Issue	Symptoms & Secondary Failures
Running demanding computer vision models alongside UI rendering in a single thread overloads the websocket.	Severe video lag, frozen parameter controls, and the dreaded white screen of death . Fragile path dependencies on Windows and unstable native hardware capture causing environment-dependent crashes.

Bifurcating the system unlocks absolute stability

The Result: A true Dual Engine system where UI elements can never be blocked by ML inference loops.



Dashboard

Live Detection

Stream from webcam, IP camera, or video file with real-time RT-DETR inference, FPS counter, per-class alert thresholds, and one-click frame capture for retraining.

 [Open Live Detection →](#)

File Detection

Upload individual images or videos, run batch inference over entire folders, download annotated outputs and export results as CSV or JSON.

 [Open File Detection →](#)

Analytics

Explore your full detection history — class distribution, confidence histograms, detections-over-time charts, and session summaries. All backed by SQLite.

 [Open Analytics →](#)

Annotation Studio

Draw bounding boxes over captured edge-case frames, assign COCO class labels, review dataset stats per class, and export your dataset as a ready-to-train ZIP archive.

 [Open Annotation Studio →](#)

Model Settings

Switch between RT-DETR model sizes, load a custom `.pt` weight file, view benchmark stats, and launch a fine-tuning run on your annotated training data.

 [Open Model Settings →](#)

MJPEG stream · zero Streamlit latency · FastAPI backend

⚡ Control

▶ Start Stream

⏏ Stop Stream

📷 Capture Frame

🔄 Apply Config

📊 Server

● LIVE

💡 CPU



Live MJPEG · direct browser connection · [open standalone](#) ↗

🔍 Stats

Inference FPS ⓘ

0.5

Capture FPS ⓘ

9.9

Frames done

10

Total dets.

17

⚡ Control

▶ Start Stream

■ Stop Stream

📷 Capture Frame

⚙️ Apply Config

📶 Server

● LIVE



Live MJPEG · direct browser connection · [open standalone ↗](#)

🔍 Stats

Inference FPS ⓘ

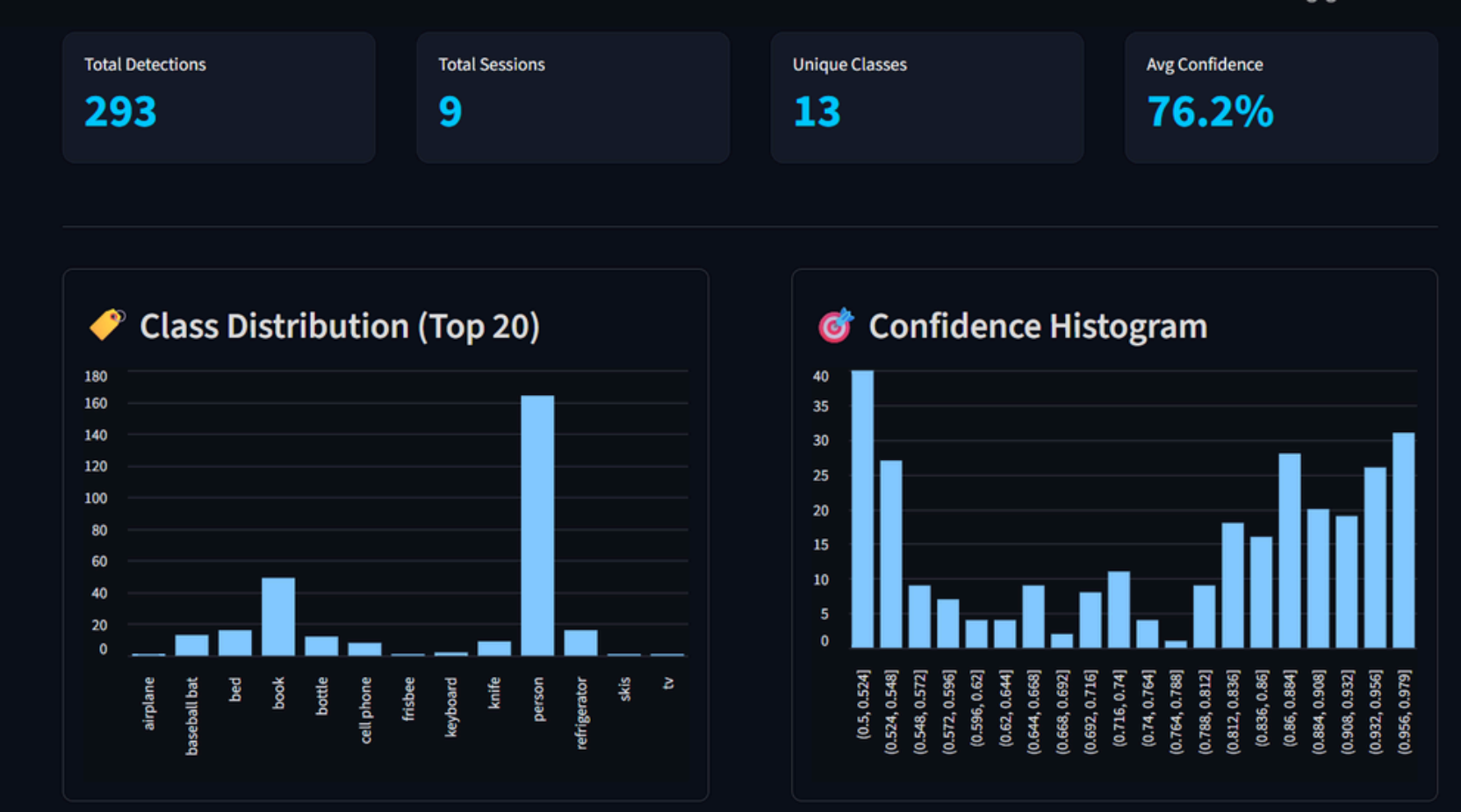
0.6

Capture FPS ⓘ

9.9

Frames done

385



 Images

Select frame

capture_202605... 

1 image(s) in dataset

 Delete This Image

 Label 

Class to annotate

person (id:0) 

Drawing mode: drag to draw a box. Canvas shows existing labels.

 Save

 Clear

A modern Python stack optimized for low-latency computer vision

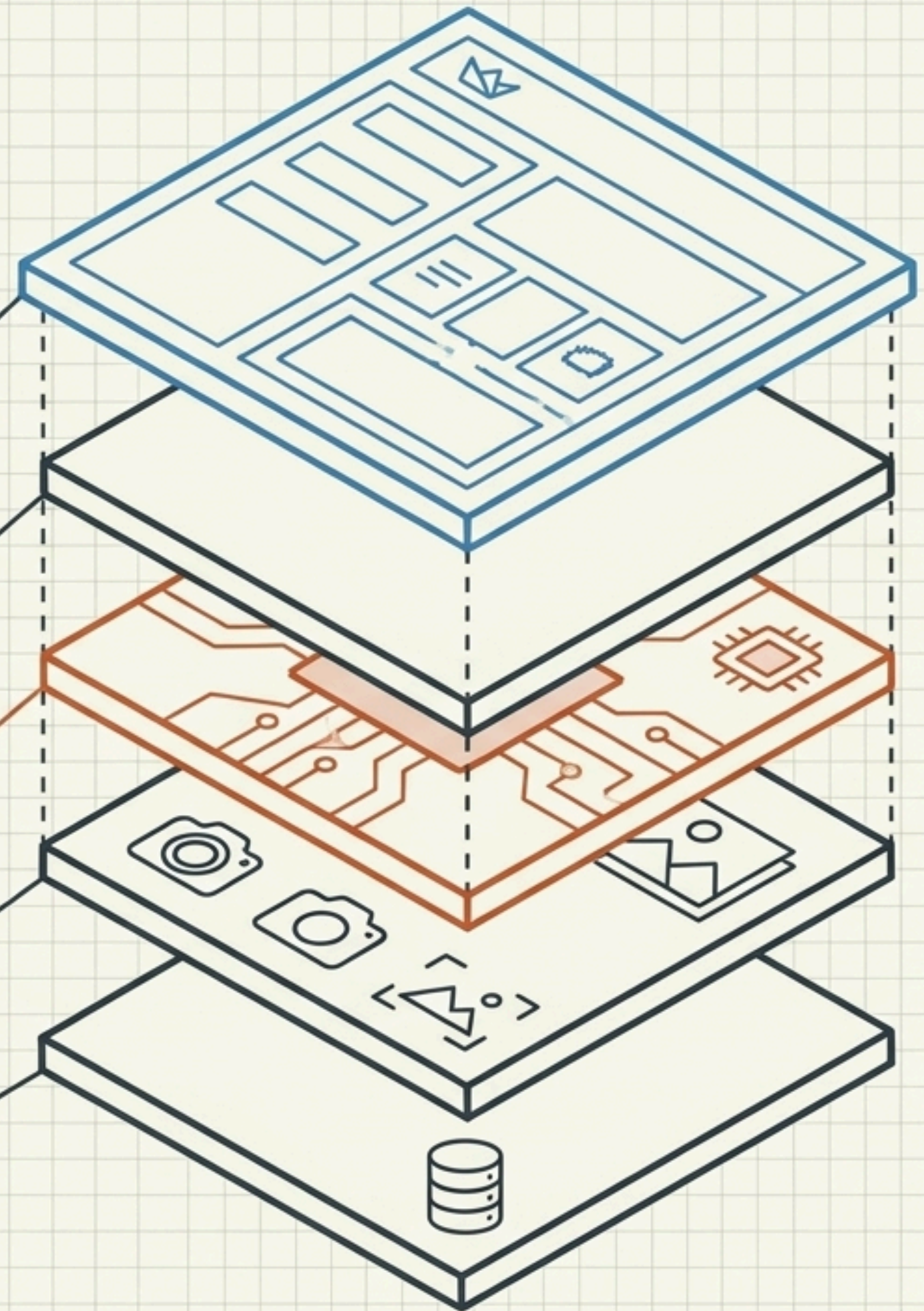
Layer 5 (Frontend): Streamlit (Component-based dashboard, interactive charting, multi-page layouts).

Layer 4 (Video Pipeline): FastAPI + Uvicorn (Asynchronous MJPEG stream serving on port 8502).

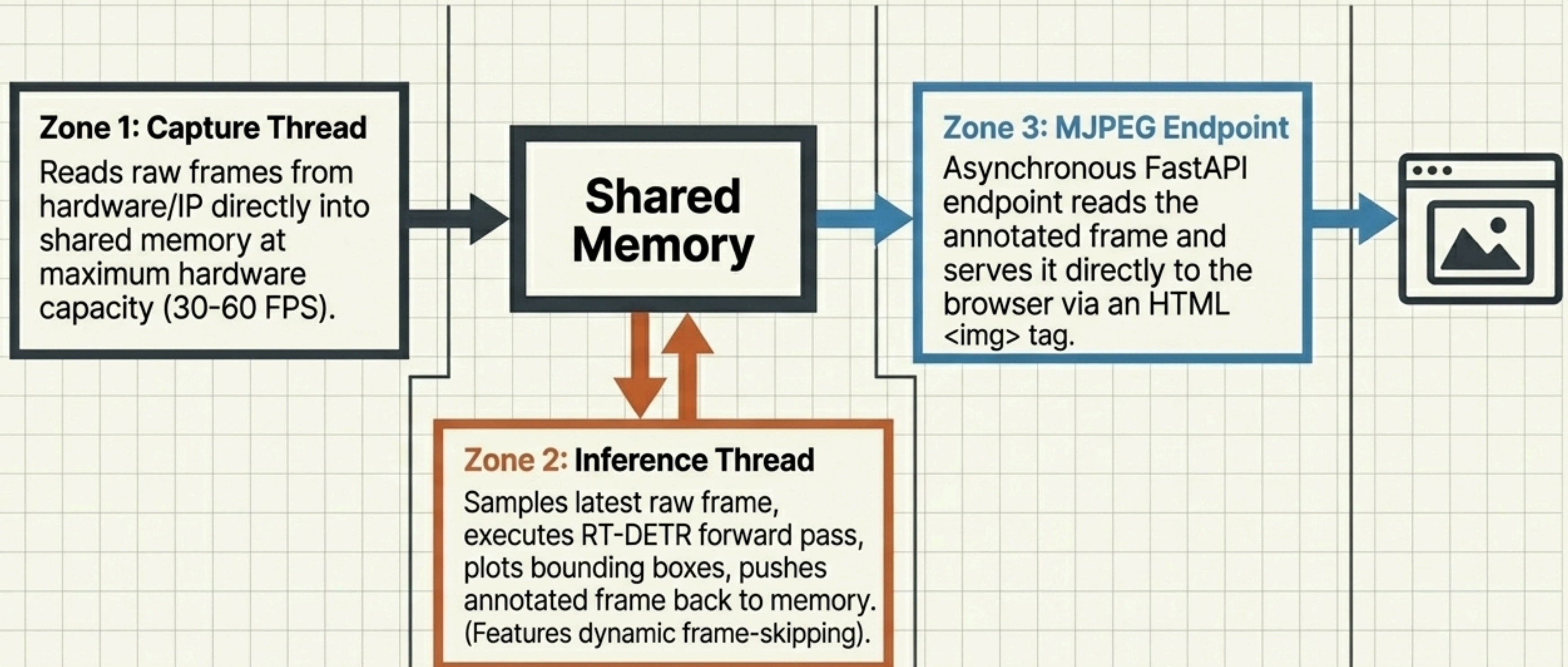
Layer 3 (Core Model): RT-DETR via Ultralytics (Transformer-based object detection).

Layer 2 (CV Processing): OpenCV / cv2 (Hardware-accelerated capture and frame encoding).

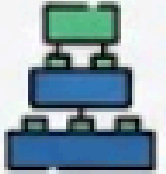
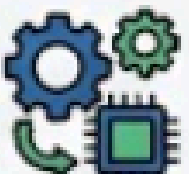
Layer 1 (Data Persistence): SQLite (Low-footprint detection logging and session summaries).



Isolating threads guarantees zero-latency frame delivery



DETR vs. RT-DETR: A FEATURE COMPARISON

FEATURE	Original DETR (2020)	RT-DETR (Modern)
 SPEED	Slow. Not suitable for real-time video.	Real-Time. 100+ FPS on high-end GPUs.
 BACKBONE	Standard ResNet.	HGNetv2 (much faster on GPUs).
 CONVERGENCE	Very Slow. Needed 500 epochs to train.	Fast. Good results in 50-100 epochs.
 ENCODER	Dense & heavy.	Hybrid Encoder (processes scales separately).
 SMALL OBJECTS	Struggled to see small things.	Improved significantly via multi-scale features.

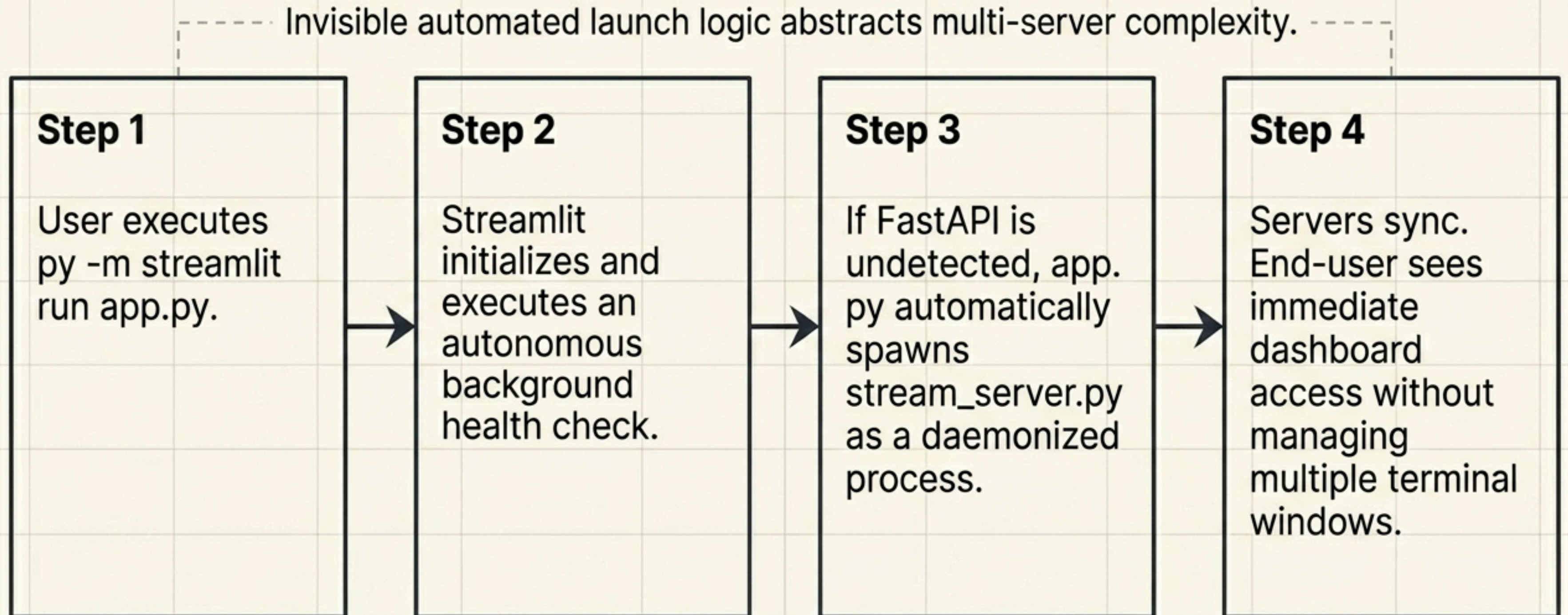
*FPS (Frames Per Second)

*Epochs (Number of full training passes through the dataset)

The decoupled architecture completely eliminates UI blocking

	Monolithic Architecture	RT-DETR Sentinel Pro
UI Stability	Frequent freezing & White Screens of Death	<u>100% interactive sliders and controls during live inference</u>
Video Framerate	Severe lag tied to render loops	Native browser rendering at <u>30+ FPS</u> via <u>MJPEG</u>
System Overload	Queue backlog leading to eventual crash	Graceful CPU Degradation (N-frame skip keeps live feed smooth while inference throttles)

Daemonized polling creates a single-command launch experience



Technical optimizations unlock distinct operational capabilities

Edge-Device Readiness

Enabled by the decoupled N-frame skip mechanism. The suite runs smoothly on non-CUDA hardware (Intel NUCs, Raspberry Pi) by throttling inference while keeping human-operator video feeds flawlessly smooth.

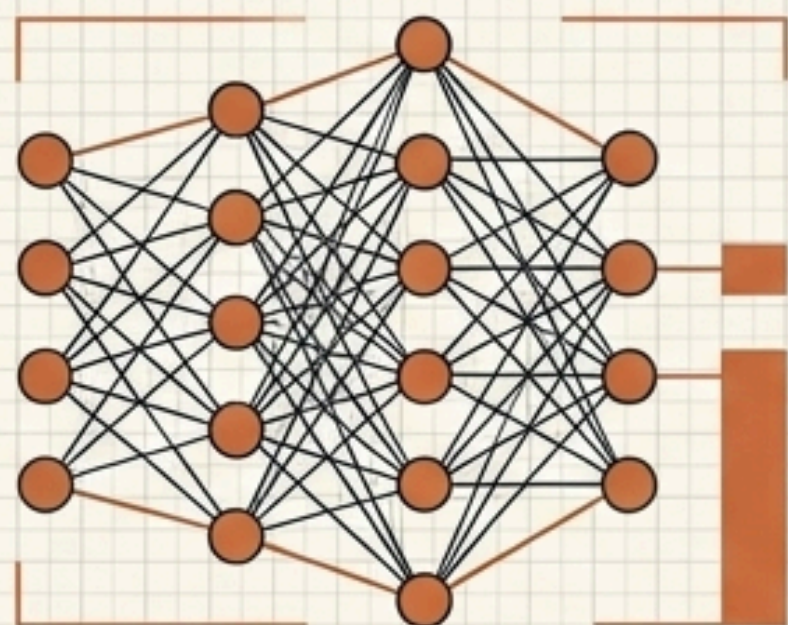
Active Learning Loop

Enabled by the unblocked UI. Floor operators can use the Annotation Studio and Capture Frame tools in real-time to isolate edge cases and export YOLO-format data for continuous fine-tuning.

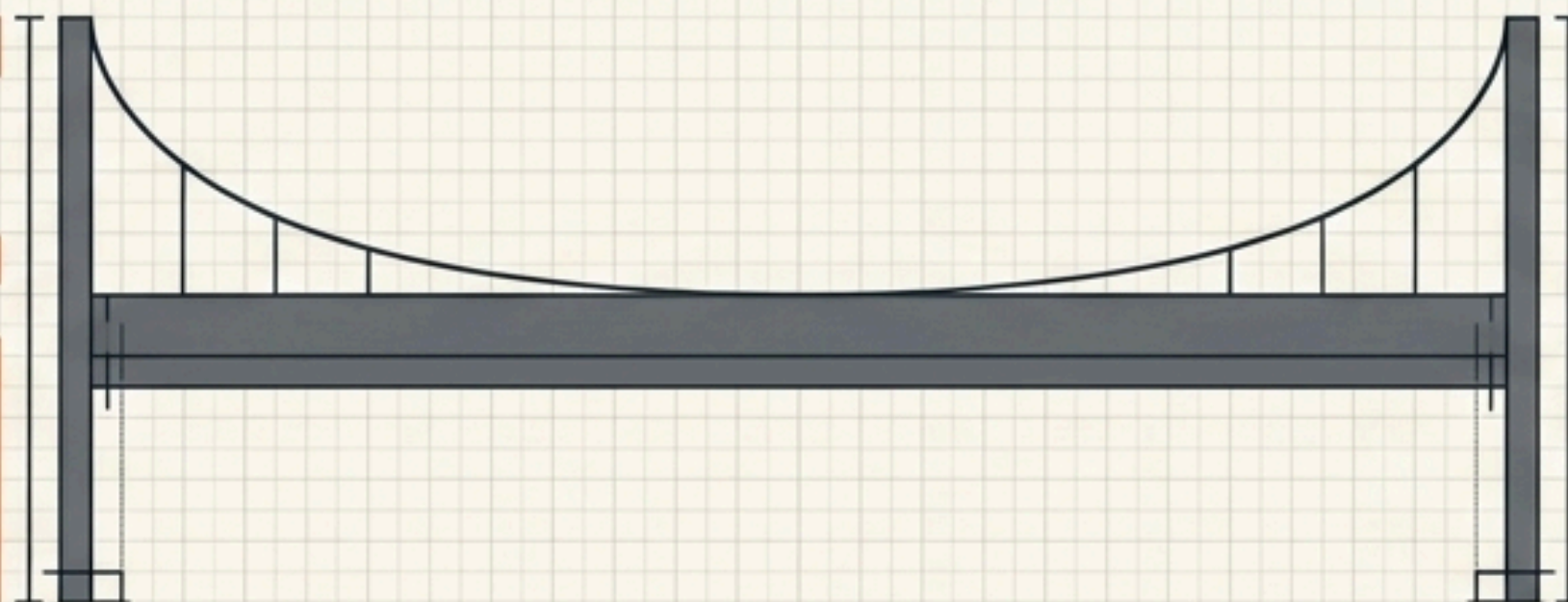
Agnostic Sourcing

Enabled by OpenCV fallback chains. Instant deployability across RTSP IP cameras, direct webcams, and pre-recorded loop files for immediate retail or security analytics.

Delivering enterprise-grade computer vision without compromise



SOTA Models



Dashboard Analytics

Key Takeaway 1: RT-DETR

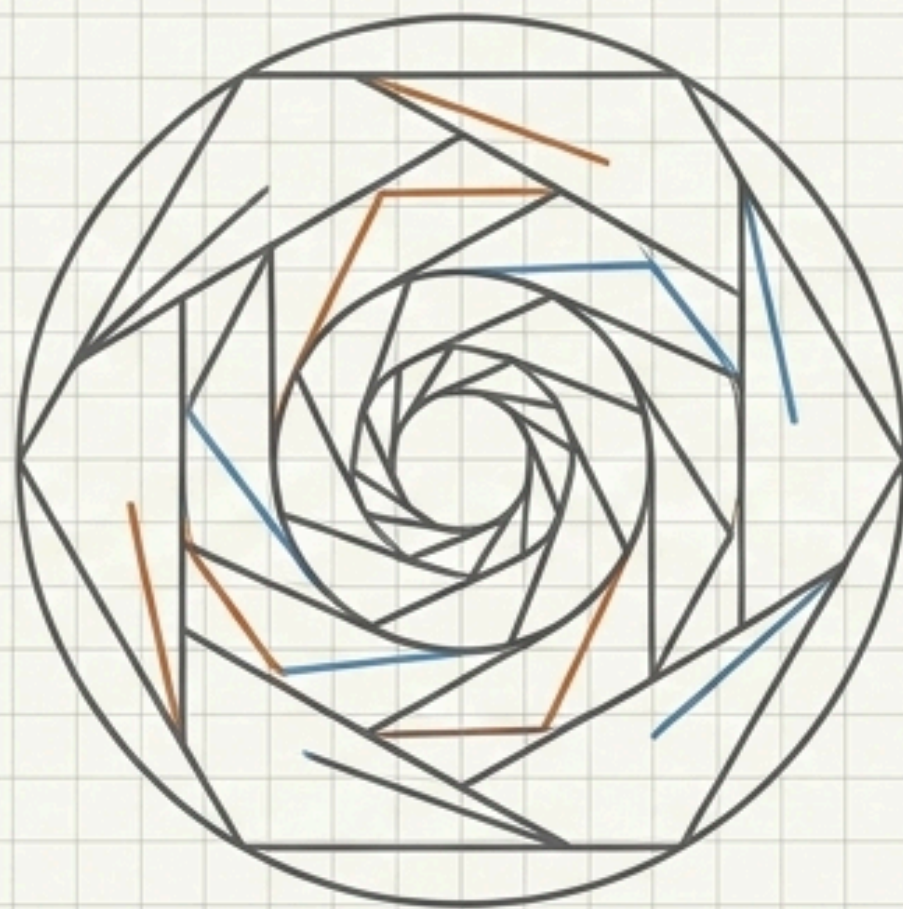
Sentinel Pro successfully bridges the gap between state-of-the-art transformer object detection and accessible, real-time analytics.

Key Takeaway 2: By abandoning

synchronous constraints for a decoupled MJPEG/FastAPI backend, the platform guarantees absolute UI stability under massive computational loads.

Key Takeaway 3: Integrated

auto-recovery and path-handling result in an end-to-end, highly optimized solution ready for immediate enterprise deployment.



Thank You

RT-DETR Sentinel Pro Technical Report